

Jet Chang

+1 (289) 218-8818 | jet.chang@mail.utoronto.ca | <https://jcj1337.github.io/Website> | github.com/jcj1337

Education

University of Toronto, BAsC. in Industrial Engineering + PEY Co-op

Expected 2028

- Minor in Artificial Intelligence
- Relevant Coursework: Data Science, Data Modelling, Machine Learning, Probability, Statistics, Object-Oriented Programming, Systems Modelling & Simulations, Operations Research I, II, & III, Data Structures & Algorithms, Human Factors

Skills

Programming Languages: Python, SQL, Java, R, MATLAB, HTML, CSS

Libraries/Tools: pandas, NumPy, scikit-learn, PyTorch, XGBoost, SciPy, Matplotlib, Tableau, Power BI, Jupyter, Git, AWS, React, REST APIs, GraphQL, Docker, Streamlit, Spring Boot, gurobipy, Simio, Appsmith, Excel

Experience/Projects

Pre-Game Win Probability Estimator | [Github](#)

Jan. 2026 - Apr 2026

University of Toronto

- Built and tuned a **3-layer transformer-based pre-game win probability model** with a learned **CLS token** to summarize interactions across the full draft, then combined the resulting draft-aware representation with player profile features in a final **MLP** head
- Designed an **end-to-end** data pipeline for API collection (**200k+ multi-region samples**), feature engineering, and held-out evaluation using log loss, expected calibration error (**~0.5%**) and Brier score
- Outperformed best tuned logistic regression and XGBoost models, achieving **~72.2%** and **~37.5%** less calibration error respectively
- Developed engineered draft and player-summary features that improved log loss by **~5%**
- Added a **parallelized** multi-region API ingestion pipeline that collected ranked match data concurrently across regions, for **4x** throughput.
- Implemented caching for repeated API lookups, reducing dataset build time during iterative experimentation by **~52%** on subsequent runs

Commodities Trading Backtest

- Built a rule-based commodities backtesting pipeline for Brent and WTI crude oil using trend/regime signals, volatility targeting, and transaction-cost-aware evaluation over **20+** years of data
- Improved out-of-sample Brent strategy performance to **66.6%** total return (**0.57 Sharpe**) after adding **volatility targeting** and **regime continuation** versus a negative-return baseline
- Designed a **46/54** inverse-volatility allocation scheme to balance portfolio risk, improving diversification relative to a single-asset strategy
- Published an interactive **Tableau** dashboard to visualize equity curves, positions, and drawdowns, making results easier to interpret and compare | [Dashboard](#)

Manga Recommendation System Backend

- Implemented a data pipeline for manga metadata ingestion, cleaning, and feature engineering, storing **raw and transformed datasets in AWS S3** for reproducible downstream analysis
- Used **GraphQL** to extract nested manga metadata efficiently from upstream sources and transform it into feature-ready tables using NumPy
- Collaborated in a 2-person team using Git for version control and branch-based development

Reinforcement Learning Strategy Simulation | [Github](#)

- Modelled interactions in a competitive multiplayer environment as a **2-agent** Markov decision process
- Implemented a **Q-learning** agent trained against a rotating pool of baseline opponent policies, with periodic snapshotting and fixed-policy evaluation to measure policy robustness across matchups over **5000+** training episodes
- Achieved **88%–89% win rate** in simulation and learned interpretable, state-dependent strategies aligned with **domain intuition**

Additional Extracurriculars:

- Independent eSports Coach 2022-2024
 - Dozens of clients and consistent rank improvements, articulated decision-making under uncertainty real-time
- Top 50 (0.0003%) in Ranked Solo/Duo League of Legends